European Polymer Journal 220 (2024) 113463

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

Efficient and facile dual post-polymerization modification using acetoacetate-acrylate Michael addition

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ARTICLE INFO

Keywords:
β-ketoesters
Post-polymerization modification
Michael addition
Acetoacetate

ABSTRACT

Post-polymerization modification (PPM) is a valuable strategy for achieving diversified functional polymers; nonetheless, the pursuit of achieving mild and efficient dual modification along the polymeric backbone remains as a challenge. In this work, the Michael addition reaction of acetoacetates and acrylates have been demonstrated as a robust PPM technique. The reversible addition-fragmentation chain transfer polymerization of commercially available 2-(acetoacetoxy)ethyl methacrylate provides a polymer scaffold with controlled molecular weight and acetoacetate groups. These pendant groups are capable of undergoing efficient double addition reactions with acrylates, leading to an increased density of functional moieties within the polymeric backbone. This PPM reaction exhibits unique advantages such as mild reaction conditions (at 40 °C), rapid kinetics (completion within 1 h), and high atom economy. Consequently, a series of polymers with tunable thermal stability, glass transition temperature, and hydrophilicity have been successfully synthesized. Applications have been demonstrated for both the synthesis of polymers exhibiting aggregation-induced luminescence and the preparation of luminescent materials with adjustable mechanical properties. This pioneering work unveils a robust post-polymerization modification technique, significantly advancing the synthesis of innovative functional polymers.

1. Introduction

Advanced polymers featuring a myriad of functional groups are increasingly needed in various interdisciplinary areas like personalized healthcare, diagnostics, cutting-edge electronics, and high-end structural materials [1,2]. To meet the demand for structural diversification, post-polymerization modification has become one of the most valuable tools in polymer chemistry [3–5], which entails the chemical transformation of reactive polymer precursors and precision tuning of architectures and properties. Starting from a single reactive polymer precursor, the post-polymerization modification approach enables the generation of a diverse library of functional polymers with identical average chain lengths and chain-length distributions [6], making it particularly appealing for combinatorial materials exploration and the establishment of structure–property correlations. Successive polymerization and modification can also avoid the limitation of polymerization mechanism on the compatibility of functional groups [7]. The rapid advancement of controlled polymerizations with higher functional-group tolerance has further propelled the thriving progress of post-polymerization modifications, offering opportunities to access complex polymer structures that were once deemed unattainable [8–10].

The ideal post-polymerization modification reactions should be mild, rapid, efficient, no side reactions, possess high functional group tolerance, and utilize readily available substrates [11]. Over the past decade, the emergence of click chemistry has introduced a variety of reactions for modifying polymers, including copper-catalyzed azide-alkyne cycloaddition reaction [12,13], activated ester-amine reaction [14], Diels-Alder cycloaddition reaction [15,16], thiol-ene/yne reaction [17–20], amine/thiol-epoxy reaction [21,22] and others. Multicomponent reactions like Kabachnik-Fields [23,24], Ugi [25–27], and Biginelli reactions [28–30] have been used for dual modification of polymers. Despite these progresses, most reactions face challenges such as the requirement for metal catalysis, substrate sensitivity (e.g., active esters prone to hydrolysis, thiols susceptible to oxidation, and azides being explosive) or limited accessibility of substrates.

β-ketoesters, including acetoacetates, malonates, and 1,3-diketones, demonstrate keto/enol tautomerism and exhibit both electrophilic and nucleophilic functionalities, rendering them a fundamentally important building block in organic chemistry [31]. The reactions between β-ketoesters and amines yield exchangeable vinylogous urethanes,

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https://doi.org/10.1016/j.eurpolymj.2024.113463
Received 19 June 2024; Received in revised form 7 September 2024; Accepted 13 September 2024
Available online 21 September 2024
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utilized for reversible polymer modification [7] and dynamic covalent material preparation [32,33]. The reversible Knoevenagel condensations of β-ketoesters and aldehydes are versatile for reversible modifications [34] and vitrimeric material preparation [35,36]. β-ketoesters participate in multi-component reactions such as Hantzsch [37,38] and Biginelli reactions [39,40], playing a key role in the design and synthesis of complex polymers through multi-component polymerization strategies. Furthermore, β-ketoesters undergo Michael addition reactions with acrylates, valuable for preparing cross-linked materials [41–43].

To further enrich the post-polymerization toolbox, and inspired by β-ketoester chemistry, we explored the Michael addition reaction of acetoacetates as a new and facile tool for highly functionalized polymers (Scheme 1). As far as we know, the utilization of this efficient reaction in PPM has not yet been reported. The reversible addition-fragmentation chain transfer (RAFT) polymerization of commercially available 2-(acetoacetoxy)ethyl methacrylate (AAM) provides a polymer scaffold with controlled molecular weight and acetoacetate groups. These pendant groups are capable of undergoing efficient double addition reactions with acrylates, leading to an increased density of functional moieties within the polymer backbone. The PPM process is characterized by its mild reaction conditions (conducted at 40 °C), rapid kinetics (completed within 1 h). Through strategic selection of the incorporated functional groups, the thermal stability, glass transition temperature, and hydrophilicity of the resultant polymers can be finely tuned. This PPM approach has been leveraged to synthesize polymers exhibiting aggregation-induced luminescence (AIE) and produce photoluminescence polymer materials with customizable mechanical characteristics.

2. Results and discussion

2.1. Synthesis of poly(2-(methacryloyloxy)ethyl acetoacetate)

The poly(2-(methacryloyloxy)ethyl acetoacetate) (PAAM) containing β-ketoesters was selected as the polymer backbone to explore PPM via the Michael addition reaction. According to the literature [44], this polymer was synthesized by RAFT polymerization of 2-(methacryloyloxy)ethyl acetoacetate (AAM), where 2-dodecylsulfanylcarbono-thioylsulfanyl-2-methylpropanoic acid (DSCTSPA) served as the chain transfer agent and AIBN was used as the initiator. The molar ratio AAEMA:DSCTSPA:AIBN was set to 50:1:0.1. After polymerization at 70 °C for 12 h, the resulting PAAM polymer was purified by precipitation in methanol. The ^{1}H NMR spectrum shown in **Figure S6** revealed its clear chemical structure, with characteristic acetoacetate ester group signals at 3.56 and 2.29 ppm and the intact RAFT end group at 1.2 ppm. GPC analysis indicated that the polymer had a high M_n of 15,800 g/mol and a dispersity of 1.45. TGA analysis demonstrated that the polymer exhibited good thermal stability, with a 5 % weight loss temperature ($T_{d,5\%}$) reaching 196 °C and complete decomposition occurring at 500 °C. DSC measurement showed that the polymer was amorphous with a glass transition temperature (T_g) at −4 °C.

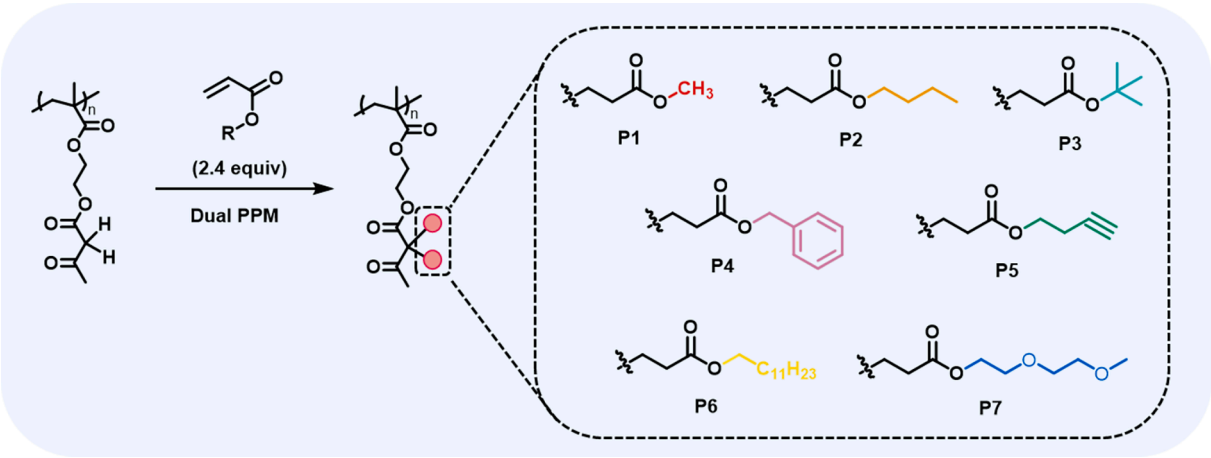
2.2. Demonstration of the PPM reaction

With this polymer in hand, we investigated its PPM via the Michael addition of β-ketoesters with acrylates. Specifically, PAAM was modified using a model compound methyl acrylate to give polymer **P1**. To facilitate the structural analysis, the reaction of ethyl acetoacetate and methyl acrylate was performed to produce the model addition compound **A1**. In principle, the active methylenes in acetoacetate groups were capable of undergoing twice addition with acrylates. Therefore, for the PPM and model compound synthesis, the molar ratio of acetoacetate to acrylate was set to 1:2.2. The Michael addition reaction was carried out in THF at 40 °C and catalyzed by 1 mol% DBU for 1 h.

Both ^{1}H NMR and ^{13}C NMR spectra of ethyl acetoacetate, **A1**, and **P1** were compared in Fig. 1. In the ^{1}H NMR spectra of **A1** and **P1**, the active methylene peak at 3.44 ppm of acetoacetate groups completely disappeared after the reaction, indicated its complete conversion. Both **A1** and **P1** exhibited characteristic methoxy group peaks f at 3.50 ppm and 3.63 ppm respectively. Moreover, the integral ratio of the methoxy peak h to the methyl peak b in both addition products was 2:1, indicating the dual addition of acetoacetate groups by methyl acrylate (**Figure S14**). The ^{13}C NMR spectra of both addition products revealed the peaks **9** at 172 ppm and **10** at 51 ppm corresponding to the carbonyl carbon and the methoxy carbon introduced by methyl acrylate and high similarity of the FTIR spectra of model compound and P1 (**Figure S1**) further supporting the occurrence of Michael addition. **Figure S4** displayed the unimodal and symmetrical GPC curve of **P1**, which shifted to a lower retention time than PAAM, indicated an increase in the polymer's molecular weight resulted from the PPM.

2.3. Optimization of PPM conditions

To further optimize the PPM reaction, we investigated the effects of key parameters such as reaction time, temperature, DBU concentration, and solvents. Benzyl acrylate was selected as the model compound to facilitate monitoring the reaction kinetics by 1H NMR analysis. In the presence of DBU, the β-ketoesters containing two active hydrogens undergo a two-step process [45] (Scheme 2). First, a carbon anion is formed, which then undergoes a single addition with the acrylate. This is followed by a second addition, leading to a dual-addition product. The ratio of dual-addition to mono-addition depends on the reaction process



Scheme 1. Dual post-polymerization modification of PAAM with various acrylates.

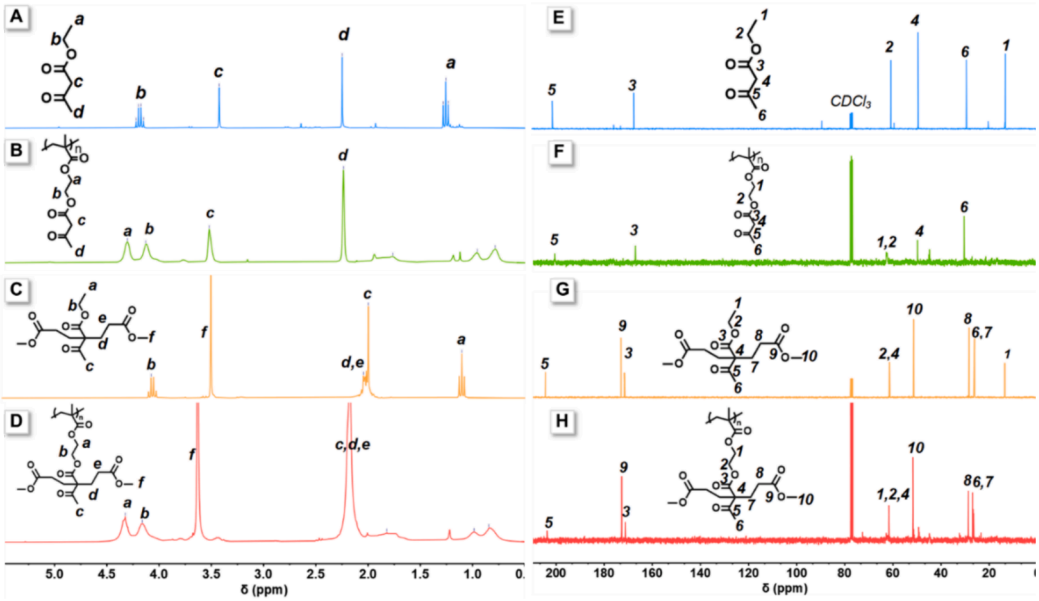
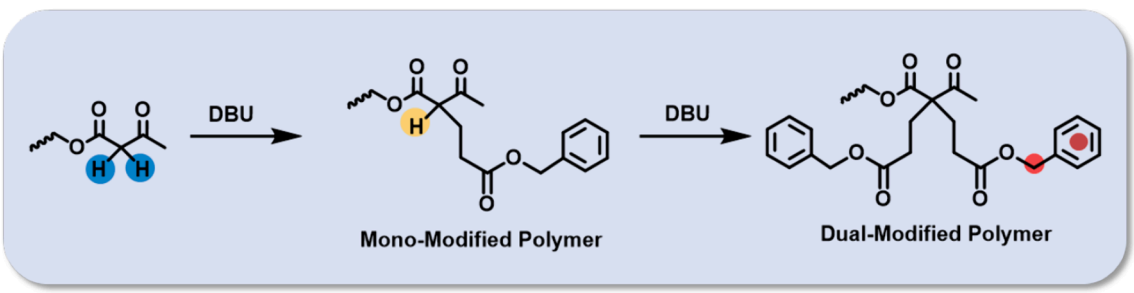


Fig. 1. ^{1}H NMR spectra of EAA(A); **PAAM**(B), **A1**(C)and **P1**(D) in chloroform-*d*. ^{13}C NMR spectra of EAA(E); **PAAM**(F), **A1**(G)and **P1**(H) in chloroform-*d*.



Scheme 2. Mechanism for the dual-modification of PAAM with benzyl acrylate.

and the stoichiometry of the reactants.

We closely monitored the reaction between PAAM and benzyl acrylate over time. The ^{1}H NMR spectra of the purified polymer products (obtained by precipitation) were shown in Fig. 2A. The signal corresponding to the active methylene group at 3.56 ppm gradually diminished, while the characteristic benzyl peaks at 7.25 ppm and 5.00 ppm intensified. Additionally, the signal attributable to the CH of the mono-addition product was detected at 3.71 ppm. The conversion of β-ketoester groups reached 36 %, 95 %, and 100 % after 10 min, 30 min, and 60 min, respectively, indicating the rapid kinetics of the Michael addition reaction (Fig. 2B). Further ^{1}H NMR analysis (**Figure S7-S13**) showed that the molar ratio of dual-addition to mono-addition increased from 0.6:1 to 7.9:1 and finally to 1:0 over the course of the reaction, suggesting the selective formation of the dual-modification product after 60 min (Table 1). The amount of DBU was found to be a crucial factor affecting the β-ketoester conversion. Increasing the DBU concentration from 0.1 % to 0.5 % to 1 % led to substantial improvements in the conversion, from 2 % to 97 % to 100 %, respectively. The suitable amount of DBU played a key role in generating the necessary carbon anions to catalyze the Michael addition reaction. Moreover, the reaction

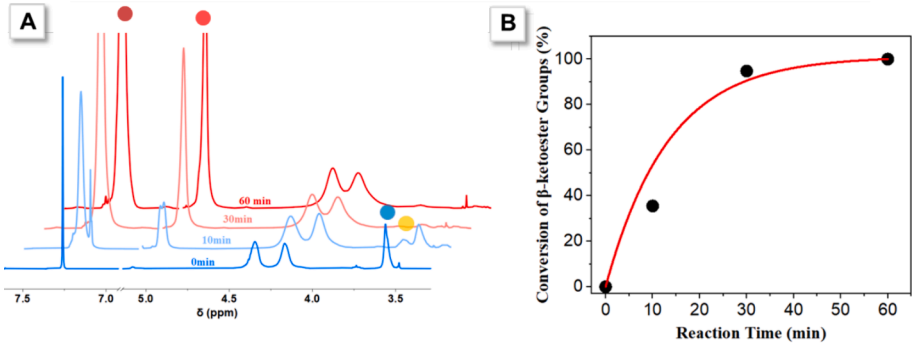


Fig. 2. Dual post-polymerization modification of PAAM with benzyl acrylate. A) Overlaid ^{1}H NMR spectra after different reaction intervals. At each time point, the polymer underwent precipitation with MTBE to effectively remove unreacted benzyl acrylate. (B) Conversion of β-ketoester groups in PAAM modified with benzyl acrylate over time.

Table 1
Effect of reaction conditions of PAAM with benzyl acrylate on the conversion of β-ketoester and molar ratio of dual-addition to mono-addition.[a]

Entry	Time(min)	Solvent	Amount of DBU (mol%)	Temperature (°C)	Conversion of β-ketoester groups[b] (%)	Dual addition: Mono-addition
1	60	THF	1	40	100 %	1:0
2	30	THF	1	40	95 %	7.9:1
3	10	THF	1	40	36 %	0.6:1
4	60	THF	0.5	40	97 %	1:0
5	60	THF	0.1	40	2 %	0:1
6	60	THF	1	30	90 %	3.8:1
7	60	THF	1	20	36 %	0.6:1
8	60	1,4-Dioxane	1	40	100 %	1:0
9	60	DMF	1	40	100 %	1:0

[a] Initial acetoacetate concentration was 0.5 mol·L^{-1}.
[b] Determined by ^{1}H NMR measurements.

temperature had a significant impact on the polymer modification efficiency. The conversion of β-ketoester groups increased from 36 % at 20 °C to 90 % at 30 °C, and eventually reached 100 % at 40 °C. In contrast, the choice of solvent (dioxane or DMF) had a minimal effect on the efficiency of the PPM process under the investigated conditions. Considering the lower boiling point of THF, it was selected as the solvent for further investigations. Based on these findings, the optimized reaction conditions were determined to be: tetrahydrofuran as the solvent, 1 mol% DBU, a temperature of 40 °C, and a reaction duration of 1 h.

2.4. Generality of dual PPM

Under the optimized modification conditions, we further extended this PPM strategy to create a library of functional polymers by incorporating various acrylates. Specifically, butyl acrylate, *tert*-butyl acrylate, butynyl acrylate, dodecyl acrylate, and methoxy diethylene glycol acrylate were employed (Fig. 3). The resulting polymer adducts, following precipitation purification, were obtained in high yields exceeding 85 %. Their modification molar ratios and conversion of β-ketoester were calculated based on the ^{1}H NMR integration ratios shown in **Figure S14-20**.

GPC results in **Figure S5** exhibited monomodal curves that shifted towards lower retention time, confirming the occurrence of the addition reactions manifested by increased molecular weights. The dispersity values of the resulting polymers remained approximately 1.45, in alignment with the precursor PAAM (**Table S1**). Additionally, FTIR spectroscopy facilitated the characterization of the produced adducts (**Figure S2**). Polymer **P4** showcased distinctive peaks at 775 cm^{-1} and 750 cm^{-1} corresponding to the phenyl ring C–H bending vibrations, while polymer **P5** displayed a characteristic vibration peak at 3300 cm^{-1} indicative of the alkyne group [46]. This PPM strategy based on PAAM enabled the synthesis of structurally diverse polymers, opening up avenues for tailoring thermal and interfacial properties. TGA analysis in Fig. 4&**Figure S25** showed that $T_{d,5\%}$ values of polymers from **P1** to **P7** exhibited apparent increases ranging from 23 °C to 77 °C compared with PAAM, indicating the.

introduction of various functional groups considerably enhanced the thermal stability. The effects on thermal-induced phase transitions varied with the kinds of acrylates for PPM. Relative to PAAM, the T_g values of polymers **P2** and **P7** exhibited a slight decline due to the introduction of butyl and methoxy diethylene glycol groups, leading to enhanced segmental mobility within the polymers (**Figure S26-27**). In contrast, polymers **P1**, **P3**, and **P4** demonstrated a modest rise in T_g values, particularly notable in polymers **P3** and **P4**, where the substantial steric hindrance from *tert*-butyl and benzyl groups impeded segmental motion, resulting in elevated T_g values. Interestingly, polymer **P6** exhibited melting behavior with the melting temperature of −34 °C, attributed to the crystalline capacity of dodecyl groups (**Figure S27**). Furthermore, the introduction of functional groups facilitates the modulation of the hydrophilicity of the polymers. Water contact angle (WCA) measurements were performed on the surface coated by various polymers (**Figure S24**). Relative to PAAM, the WCA

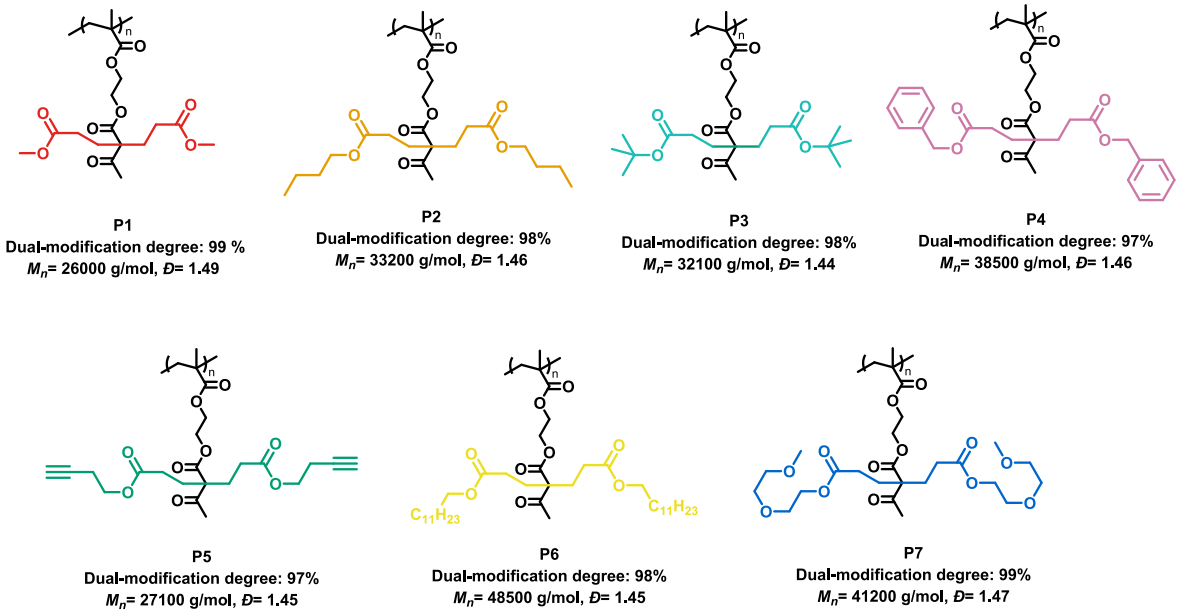


Fig. 3. Various dual-modification polymers P1-P7. Dual-modification degree, molar mass and dispersity were shown.

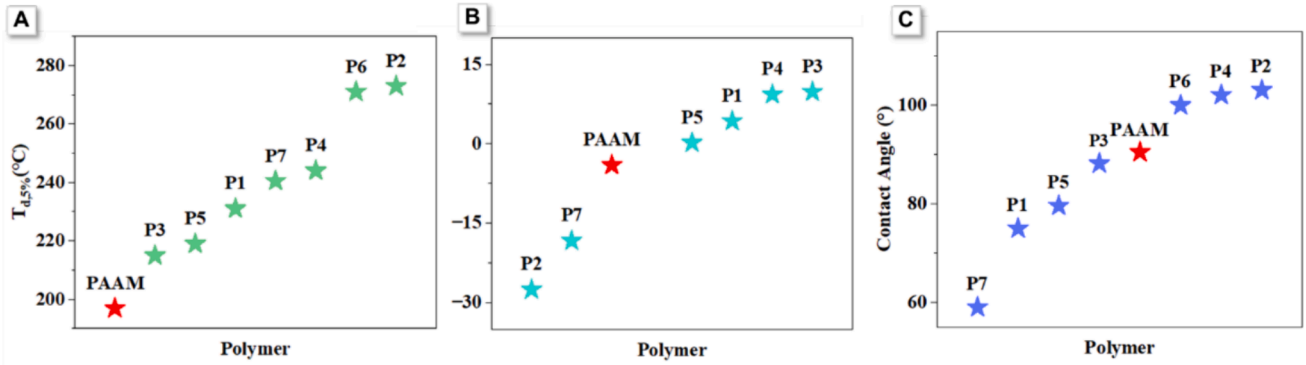


Fig. 4. (A) Scatter plot of the thermal decomposition temperatures ($T_{d,5\%}$) of polymers **P1**-**P7**. (B) Scatter plot of the glass transition temperatures (T_g) for all polymers except **P6**. (C) Scatter plot of the water contact angles of polymers **P1**-**P7**.

values of modified polymers could be flexibly adjusted from 59° to 105°. The polymer **P7** with hydrophilic ether chains, exhibited the lowest WCA value of 60°, while the polymer **P2** with hydrophobic *n*-butyl groups, demonstrated the highest WCA value of 105° (Table S2). All these results have demonstrated the effectiveness of this PPM strategy in modulating the structural and property profiles of polymers.

2.5. PPM for photoluminescent polymers.

This PPM technique allows to design polymers with specific functionality. To show this concept, tetraphenylethene-containing acrylate was utilized for the PPM of PAAM to design polymer with aggregation-induced emission (AIE) functionality.

In this reaction, 4-(1,2,2-triphenylvinyl)benzyl acrylate was used at 1 mol% relative to the β-ketoester groups. Due to the large excess of β-ketoester groups compared to the acrylate, the resulting polymer (P8) underwent mono-modification, as confirmed by the ^{1}H NMR spectra shown in Figure S21. **P8** showed strong fluorescence in the visible-light region (Figure S22). With **P8** in hand, we thus studied their emission behaviors in THF/H_2O mixtures with different water fraction ($\mathbf{f_w}$) values. As shown in Fig. 5a, weak fluorescence was detected in pure THF solution upon excitation at 348 nm. However, the emission at 470 nm was obviously enhanced with gradual addition of water into its THF solution but without changing their spectral profile. When $\mathbf{f_w}$ is 90 %, the emission of **P8** reached a maximum, and its fluorescence intensity was 7.6 times that in pure THF (Fig. 5B). With P8 in the THF/H_2O mixture at $f_w = 90$ % as an example, DLS measurement was performed to confirm its aggregation structure with the the hydrodynamic diameter of 107 nm (Figure S23). Therefore, this photoluminescence behavior confirmed that **P8** is an AIE-active polymer.

2.6. PPM for photoluminescent polymeric materials.

The highly efficient acetoacetate-based Michael addition allows not only PPM but also crosslinking of PAAM, which enables the preparation of functional polymeric materials. A crosslinked photoluminescent film was illustrated as an example. The polymer network was prepared based on the Michael addition of PAAM with both the crosslinker 1,6-hexanediol diacrylate (HDA) and the photo-luminescent 4-(1,2,2-triphenylvinyl)-benzyl acrylate.

The resulting materials were denoted as PAAM-HDA-4Pha-x (5 %, 10 %, or 20 %), where x denotes the molar percentage of HDA relative to PAAM repeat units (Scheme 3). Given the significant excess of β-ketoester groups compared to the acrylate, the mono-addition product would be the dominant reaction product. FTIR spectra was used to characterize their chemical structure. As shown in Figure S3, the C═C stretching vibration at 1630 cm^{-1} characteristic of acrylates completely disappeared, demonstrated the formation of covalently crosslinked materials via Michael addition.

The results of DSC analysis indicated that the T_g of all three materials were below room temperature. As the ratio of HDA to β-ketoester groups increased from 5 % to 20 %, the T_g of the materials increased from 9.3 °C

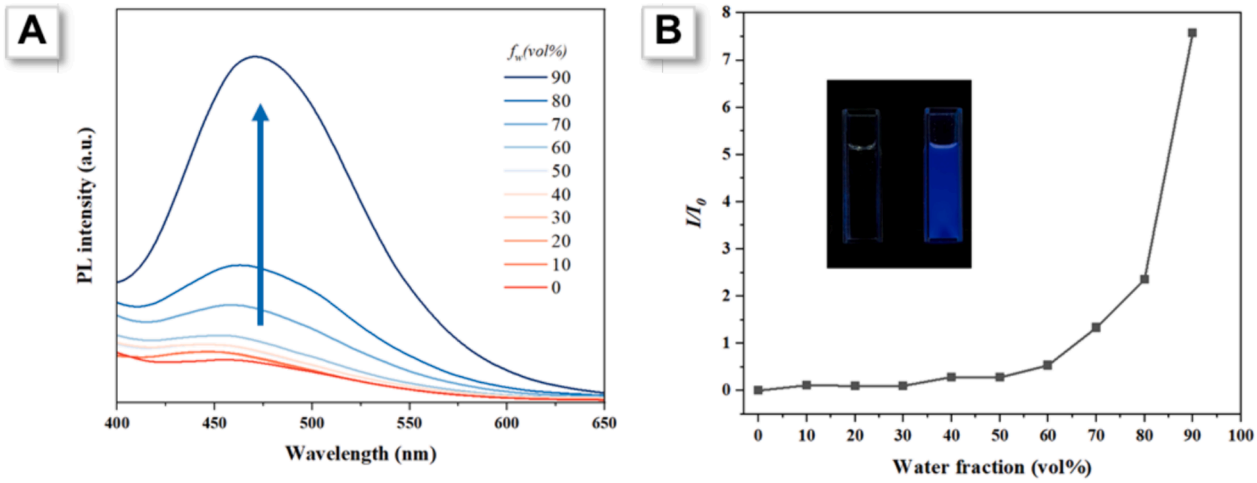
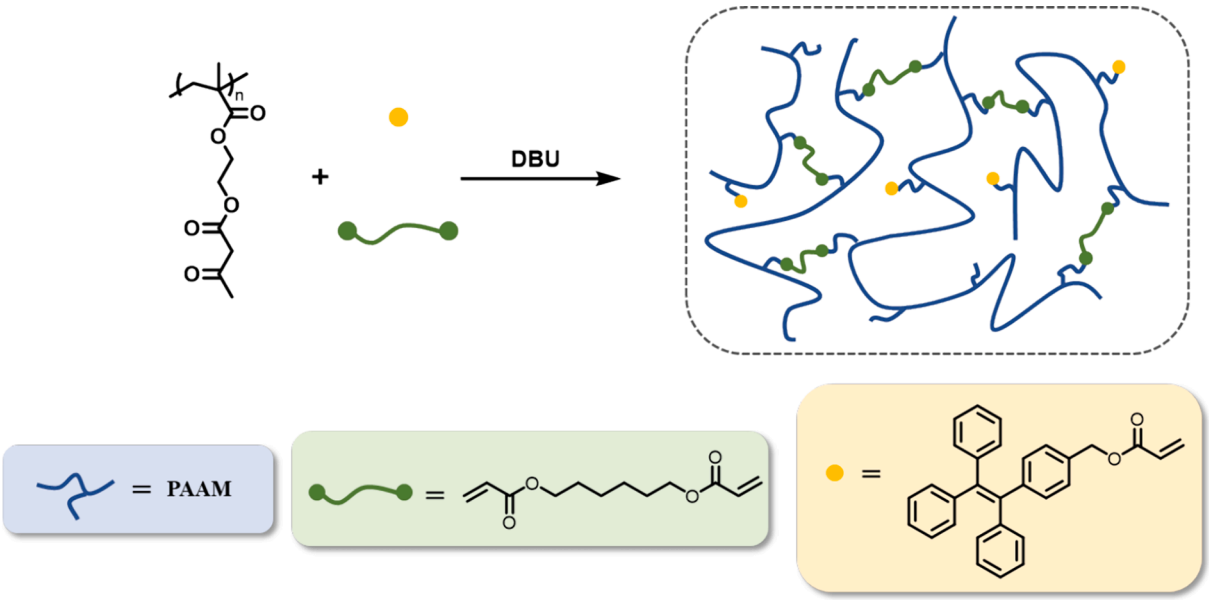


Fig. 5. (A) Fluorescent spectra of **P8** in THF and THF/water mixtures. Concentration: 1 mg/mL. λex: 348 nm. (B) Plot of relative fluorescence intensity versus water fraction in THF/water mixtures, where I is peak intensity in water mixtures and I_0 is peak intensity in pure THF. Inset in panel b: photographs of **P8** in pure THF and a THF/water mixture with 90 % water.



Scheme 3. Fabrication of photoluminescent polymeric materials.

to 18.3 °C (Figure S28). Moreover, tensile measurements of obtained samples were performed to evaluate their mechanical properties. As shown in Fig. 6a, the stress–strain curves revealed an increase in the tensile strength from 2.3 MPa for PAAM-HDA-4Pha-5 % to 9.69 MPa for PAAM-HDA-4Pha-20 %, while the fracture strain decreased from 136 % to 76 % (Table S3). The addition of more HDA increased the cross-linking degree of polymeric networks, leading to the increased tensile strength and toughness, accompanied by the decrease of tensile strain (Fig. 6B). Thanks to the AIE unit, the obtained sample exhibited fluorescent emission at 492 nm when excited at 455 nm (Fig. 6C). All these data have demonstrated that this acetoacetate-based Michael addition benefits the construction of photoluminescent crosslinked polymeric materials.

3. Conclusions

In this study, we have successfully harnessed acetoacetate-based Michael addition for the efficient and versatile post-polymerization modification of vinyl polymers. Poly(2-(acetoacetoxy)ethyl methacrylate) with controlled molecular weight and pendant acetoacetate groups are capable of undergoing efficient double addition reactions with acrylates, leading to an increased density of functional moieties within the polymer backbone. The availability of a diverse array of functional acrylates enables access to a vast library of functional materials from a single polymer feedstock, producing polymers with adjusted thermal stability, glass transition temperature, and hydrophilicity. Moreover, this PPM reaction exhibits unique advantages such as mild reaction conditions (at 40 °C), rapid kinetics (completion within 1 h), and high atom economy, exhibiting many of the hallmarks of click chemistry. Finally, we showed the application potential of this system for the synthesis of AIE-active polymers and photoluminescent soft materials. Overall, our findings highlight the significant potential of acetoacetate-based Michael addition as a post-modification technique in polymer science, paving the way toward more diverse, functional, and versatile polymer materials.

CRediT authorship contribution statement

Hongjie Chen: Writing – original draft. **Linlin Wang:** Investigation. **Shengyu Feng:** Project administration. **Lei Li:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

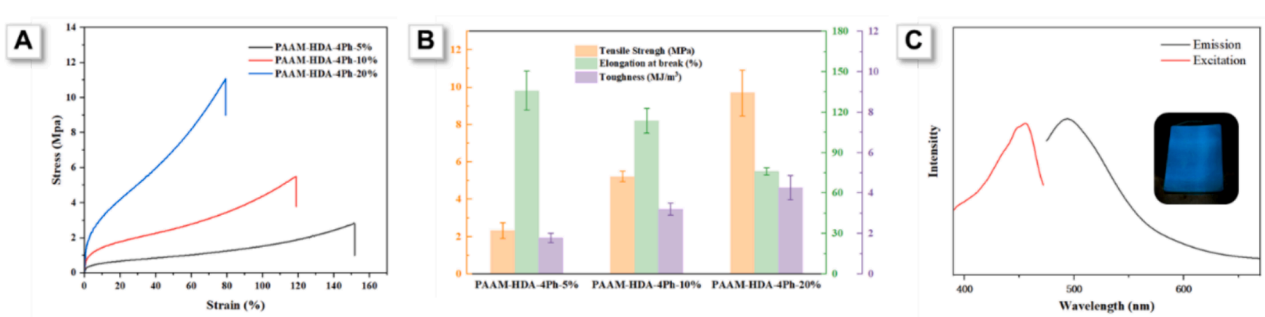


Fig. 6. (a) Stress–strain curves of samples PAAM-HDA-4Ph-5 %, PAAM-HDA-4Ph-10 % and PAAM-HDA-4Ph-20 %. (b) Tensile strength, elongation at break and toughness of samples PAAM-HDA-4Ph-5 %, PAAM-HDA-4Ph-10 % and PAAM-HDA-4Ph-20 %. (c) Excitation and emission spectra of PAAM-HDA-4Ph-10 %. Inset in panel c: photograph of PAAM-HDA-4Ph-10 % under UV light. (λem: 492 nm, λex: 455 nm).

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (No. 21901144), the Shandong Provincial Natural Science Foundation (ZR2023MB106 , ZR2020QB021), and the Young Scholars Program of Shandong University. Thermal analysis data were obtained using equipment maintained by Experimental Teaching Center at School of Chemistry and Chemical Engineering, Shandong University.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113463.

References

[1] K.A. Günay, P. Theato, H.A. Klok, Standing on the shoulders of hermann staudinger: post-polymerization modification from past to present, J. Polym. Sci. Pol. Chem. 51 (2013) 1–28. http://10.1002/pola.26333.
[2] E. Blasco, M.B. Sims, A.S. Goldmann, B.S. Sumerlin, C. Barner-Kowollik, 50th anniversary perspective: polymer functionalization, Macromolecules 50 (2017) 5215–5252. http://10.1021/acs.macromol.7b00465.
[3] A. Das, P. Theato, Activated ester containing polymers: opportunities and challenges for the design of functional macromolecules, Chem. Rev. 116 (2016) 1434–1495. http://10.1021/acs.chemrev.5b00291.
[4] Y. Gao, X. Wang, J.Z. Sun, B.Z. Tang, Synthesis of functional poly(disubstituted acetylene)s through the post-polymerization modification route, Chem. Rec. 15 (2015) 524–532. http://10.1002/tcr.201402064.
[5] Z.W. Li, H. Mutlu, P. Theato, S. Bräse, Synthesis and post-polymerization modification of poly(propargyl 2-ylidene-acetate), Eur. Polym. J. 156 (2021) 8. http://10.1016/j.eurpolymj.2021.110564.
[6] L. Yuan, L.R. He, Y.X. Wang, X.H. Lang, F. Yang, et al., Two- and three-component post-polymerization modifications based on Meldrum's acid, Macromolecules 53 (2020) 3175–3181. http://10.1021/acs.macromol.0c00482.
[7] M.B. Sims, J.J. Lessard, L. Bai, B.S. Sumerlin, Functional diversification of polymethacrylates by dynamic β-ketoester modification, Macromolecules 51 (2018) 6380–6386. http://10.1021/acs.macromol.8b01343.
[8] T. Maity, S. Paul, P. De, Designing functionalized polymers through post-polymerization modification of side chain leucine-based polymer, Eur. Polym. J. 199 (2023) 8. http://10.1016/j.eurpolymj.2023.112445.
[9] E. Akar, D. Kandemir, S. Luleburgaz, V. Kumbaraci, H. Durmaz, Efficient Post-Polymerization modification of pendant aldehyde functional polymer via reductive etherification reaction, Eur. Polym. J. 177 (2022) 8. http://10.1016/j.eurpolymj.2022.111440.
[10] N.C. Yigit, Post-modification of polyoxanorbornene via sequential "click" reactions for the preparation of flame retardant polymers, Eur. Polym. J. 207 (2024) 13. http://10.1016/j.eurpolymj.2024.112845.
[11] M.B. Larsen, S.E. Herzog, H.C. Quitter, M.A. Hillmyer, Activated polyacrylamides as versatile substrates for postpolymerization modification, ACS Macro Lett. 7 (2018) 122–126. http://10.1021/acsmacrolett.7b00896.
[12] J. Lee, S. Han, M. Kim, B.S. Kim, Anionic Polymerization of azidoalkyl glycidyl ethers and post-polymerization modification, Macromolecules 53 (2020) 355–366. http://10.1021/acs.macromol.9b02236.
[13] A. Singh, A. Agarwal, A. Chakraborty, R. Bhardwaj, S. Sutradhar, et al., Click chemistry tailored benzimidazole functionalized triazole block-co-polymer for emergence of exotic chimaeric nano-crystalsomes, Eur. Polym. J. 178 (2022) 15. http://10.1016/j.eurpolymj.2022.111503.
[14] K. Nilles, P. Theato, Synthesis and polymerization of active ester monomers based on 4-vinylbenzoic acid, Eur. Polym. J. 43 (2007) 2901–2912. http://10.1016/j.eurpolymj.2007.04.038.
[15] T.N. Gevrek, T. Bilgic, H.A. Klok, A. Sanyal, Maleimide-functionalized thiol reactive copolymer brushes: fabrication and post-polymerization modification, Macromolecules 47 (2014) 7842–7851. http://10.1021/ma5015098.
[16] K. Kempe, R. Hoogenboom, M. Jaeger, U.S. Schubert, Three-fold metal-free efficient ("Click") reactions onto a multifunctional Poly(2-oxazoline) designer scaffold, Macromolecules 44 (2011) 6424–6432. http://10.1021/ma201385k.
[17] N.K. Singha, M.I. Gibson, B.P. Koiry, M. Danial, H.A. Klok, Side-chain peptide-synthetic polymer conjugates via tandem "Ester-Amide/Thiol-Ene" post-polymerization modification of poly(pentafluorophenyl methacrylate) obtained using ATRP, Biomacromolecules 12 (2011) 2908–2913. http://10.1021/bm200469a.
[18] A.K. Williams, J. Tropp, E.R. Crater, N. Eedugurala, J.D. Azoulay, Thiol-ene click post-polymerization modification of a fluorescent conjugated polymer for parts-per-billion pyrophosphate detection in seawater, ACS Appl. Polym. Mater. 1 (2019) 309–314. http://10.1021/acsapm.8b00064.
[19] M. Basko, M. Bednarek, L.T.T. Nguyen, P. Kubisa, F. Du Prez, Functionalization of polyurethanes by incorporation of alkyne side-groups to oligodiols and subsequent thiol-yne post-modification, Eur. Polym. J. 49 (2013) 3573–3581. http://10.1016/j.eurpolymj.2013.07.018.
[20] C. Sardo, B. Nottelet, D. Triolo, G. Giammona, X. Garric, et al., When functionalization of PLA surfaces meets thiol-yne photochemistry: case study with antibacterial polyaspartamide derivatives, Biomacromolecules 15 (2014) 4351–4362. http://10.1021/bm5013772.
[21] H.B. Feng, C.G. Chae, C. Eom, G.S.W. Craig, S.J. Rowan, et al., Synthesis and characterization of amine-epoxy-functionalized polystyrene-block-poly(glycidyl methacrylate) to manage morphology and covarying properties for self-assembly, Macromolecules 56 (2023) 2675–2685. http://10.1021/acs.macromol.2c02552.
[22] M.C. Stuparu, A. Khan, Thiol-epoxy "click" chemistry: application in preparation and postpolymerization modification of polymers, J. Polym. Sci. Pol. Chem. 54 (2016) 3057–3070. http://10.1002/pola.28195.
[23] R. Kakuchi, P. Theato, Efficient multicomponent postpolymerization modification based on kabachnik-fields reaction, ACS Macro Lett. 3 (2014) 329–332. http://10.1021/mz500139c.
[24] X.Z. He, Y. Zeng, G.Q. Liu, Y. Tian, Y. Wei, et al., Magnetic self-healing hydrogel from difunctional polymers prepared via the kabachnik-fields reaction, ACS Macro Lett. 11 (2022) 39–45. http://10.1021/acsmacrolett.1c00720.
[25] L.L. Xue, X.H. Xiong, K. Chen, Y.F. Luan, G.J. Chen, et al., Modular synthesis of glycopolymers with well-defined sugar units in the side chain Ugi reaction and click chemistry: hetero homo, Polym. Chem. 7 (2016) 4263–4271. http://10.1039/c6py00734a.
[26] Y. Zeng, Y.S. Li, G.Q. Liu, Y. Wei, Y.W. Wu, et al., Antibacterial self-healing hydrogel via the Ugi reaction, ACS Appl. Polym. Mater. 2 (2020) 404–410. http://10.1021/acsapm.9b00874.
[27] A. Sehlinger, B. Verbraeken, M.A.R. Meier, R. Hoogenboom, Versatile side chain modification isocyanide-based multicomponent reactions: tuning the LCST of poly (2-oxazoline)s, Polym. Chem. 6 (2015) 3828–3836. http://10.1039/c5py00392j.
[28] Z.L. Wang, Y. Yu, Y.S. Li, L. Yang, Y. Zhao, et al., Post-polymerization modification the Biginelli reaction to prepare water-soluble polymer adhesives, Polym. Chem. 8 (2017) 5490–5495. http://10.1039/c7py01163f.
[29] C.Y. Zhu, B. Yang, Y.A. Zhao, C.K. Fu, L. Tao, et al., A new insight into the Biginelli reaction: the dawn of multicomponent click chemistry? Polym. Chem. 4 (2013) 5395–5400. http://10.1039/c3py00553d.
[30] M.Z. Yuan, H.C. Liu, K.Y. Shen, C.J. Qiu, H.S. Qi, Transparent cellulose-based film with high UV-blocking performance fabricated by surface modification using biginelli reaction, Macromol. Rapid Commun. 43 (2022) 8. http://10.1002/marc.202200495.
[31] T. Govender, P.I. Arvidsson, G.E.M. Maguire, H.G. Kruger, T. Naicker, Enantioselective organocatalyzed transformations of β-ketoesters, Chem. Rev. 116 (2016) 9375–9437. http://10.1021/acs.chemrev.6b00156.
[32] Z.Y. Liu, R. Shen, Y. Ren, X. Yang, Z.X. Shi, et al., Recyclable, malleable, tunable cross-linked elastomers based on boroxines and acetoacetyl, Eur. Polym. J. 182 (2023) 9. http://10.1016/j.eurpolymj.2022.111736.
[33] J. Bai, Z.X. Shi, Macromolecular metabolism based on enaminoneamide achieving transformation of polymer architecture, Chem. Mat. (2022) 10, http://10.1021/acs.chemmater.2c01129.
[34] L. Yuan, L.R. He, X.L. Zhang, J.H. Liu, D. Zhang, et al., Dynamic postpolymerization modification based on knoevenagel adducts of Meldrum's Acid, Macromolecules 55 (2022) 6102–6109. http://10.1021/acs.macromol.2c00422.
[35] S. Wang, H.Z. Feng, B.F. Li, J.Y.C. Lim, W. Rusli, et al., Knoevenagel C=C metathesis enabled glassy vitrimers with high rigidity, roughness, and malleability, J. Am. Chem. Soc. 146 (2024) 16112–16118. http://10.1021/jacs.4c03503.
[36] Y.W. Zhu, T.T. Man, Y.C. Tian, X.N. Zhang, J.Y. Liu, et al., Dynamic C= C Bond-based recyclable thermosetting polymers formed by knoevenagel condensation, Macromolecules 57 (2024) 1962–1969. http://10.1021/acs.macromol.3c02351.
[37] J. De-La-Cuesta, E. Verde-Sesto, A. Arbe, J.A. Pomposo, Self-reporting of folding and aggregation by orthogonal hantzsch luminophores within a single polymer chain, Angew. Chem.-Int. Edit. 60 (2021) 3534–3539. http://10.1002/anie.202013932.
[38] S. Saxer, C. Marestin, R. Mercier, Synthesis of poly(pyrazolopyridine)s by Hantzsch multicomponent polymerization, Polymer 252 (2022) 9. http://10.1016/j.polymer.2022.124935.
[39] A.C. Boukis, M.A.R. Meier, Data storage in sequence-defined macromolecules multicomponent reactions, Eur. Polym. J. 104 (2018) 32–38. http://10.1016/j.eurpolymj.2018.04.038.
[40] H.D. Xue, Y. Zhao, H.B. Wu, Z.L. Wang, B. Yang, et al., Multicomponent combinatorial polymerization via the Biginelli reaction, J. Am. Chem. Soc. 138 (2016) 8690–8693. http://10.1021/jacs.6b04425.
[41] A.O. Konuray, A. Ruiz, J.M. Morancho, J.M. Salla, X. Fernández-Francos, et al., Sequential dual curing by selective Michael addition and free radical polymerization of acetoacetate-acrylate-methacrylate mixtures, Eur. Polym. J. 98 (2018) 39–46. http://10.1016/j.eurpolymj.2017.11.003.
[42] H.Q. Zuo, Z.Y. Cao, J.B. Shu, D.D. Xu, J. Zhong, et al., Effect of structure on the properties of ambient-cured coating films prepared via a Michael addition reaction based on an acetoacetate-modified castor oil prepared by thiol-ene coupling, Prog. Org. Coat. 135 (2019) 27–33. http://10.1016/j.porgcoat.2019.05.032.
[43] N. Naga, M. Satoh, T. Magara, K. Ahmed, T. Nakano, Synthesis of porous polymers by means of Michael addition reaction of multifunctional acetoacetate and poly (ethylene glycol) diacrylate, Eur. Polym. J. 162 (2022) 9. http://10.1016/j.eurpolymj.2021.110901.

[44] Q.D. Zhang, Y.L. Zhang, Y. Zhao, B. Yang, C.K. Fu, et al., Multicomponent polymerization system combining hantzsch reaction and reversible addition-fragmentation chain transfer to efficiently synthesize well-defined poly(1,4-dihydropyridine)s, ACS Macro Lett. 4 (2015) 128–132. http://10.1021/mz500734c.

[45] S.R. Williams, K.M. Miller, T.E. Long, Michael addition reaction kinetics of acetoacetates and acrylates for the formation of polymeric networks, Prog. React. Kinet. Mech. 32 (2007) 165–194. http://10.3184/146867807x247730.

[46] S.Y. Xue, X.F. Lei, Y.Y. Xiao, G. Xiong, R.H. Lian, et al., Highly refractive polyimides derived from efficient catalyst-free thiol-yne click polymerization, Macromolecules 54 (2021) 11256–11268. http://10.1021/acs.macromol.1c01959.